CASE STUDY

System Analysis and Design for the Municipal Disaster Risk Reduction and Management Office (DRRMO)

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System Analysis and Design

ORGANIZATION DESCRIPTION

The Municipal Disaster Risk Reduction and Management Office (DRRMO) is a local government unit (LGU) tasked with disaster preparedness, response, mitigation, and rehabilitation in its municipality. Its operations include hazard mapping, risk assessment, coordination during emergencies, and relief distribution [1].

CURRENT SYSTEM

The DRRMO currently relies on manual forms, spreadsheets, and messaging apps to collect and report disaster data. These fragmented tools result in:

- Delayed information sharing and reporting,
- Lack of centralized records for incidents and resources,
- Difficulty consolidating hazard maps and response data.

This situation limits real-time coordination between municipal offices and barangay disaster teams [2](p. 21–24).

RATIONALE FOR A NEW SYSTEM

A **Local DRRM Information System** is needed to unify data management, improve communication, and support decision-making. The system will consolidate hazard data, automate reporting to higher DRR councils, and streamline response coordination [2](p. 27–29).

REQUIREMENTS

Functional Requirements

F1. User authentication and roles – Different access levels for administrators, planners, and responders.

F2. Incident and event logging – Record incidents with geotags, photos, timestamps, and update lifecycle status.

F3. Resource inventory and tracking – Manage equipment, vehicles, relief stocks, and allocation during response.

F4. Evacuation center monitoring – Register centers, track capacities, and log evacuees.

F5. Situational dashboard and maps – Show real-time hazard overlays, incident locations, and active resource status [2](p. 30–31).

F6. Automated reporting – Generate reports following NDRRMC and LGU templates.

F7. Alert and notification system – Send updates via SMS, email, and in-app alerts.

Non-Functional Requirements

N1. Availability – System must meet 99% uptime in normal periods and be designed for degraded operation (offline mobile sync) during network outages [7].

N2. Security – Role-based access, data encryption, and audit logs.

N3. Usability – User interface must be usable by non-technical staff (simple forms), support Filipino and English, and be mobile friendly for field responders.

CHOSEN GATHERING TECHNIQUES

Structured and Semi-Structured Interviews

Allows deep understanding of real-world workflows, local challenges, and reporting pain points. The DRRMO head, operations officer, and barangay captains can explain gaps in current coordination [1].

Requirements Workshops

A facilitated workshop gathers multiple stakeholders to align priorities, co-design major workflows (incident lifecycle, evacuation flow), and negotiate conflicting needs. Workshops produce consensus, accelerate validation of use cases, and generate artifacts (process maps, priority lists) that can be rapidly turned into functional requirements. Workshops are especially useful where coordination across agencies is crucial [5](p. 13–17).

REQUIREMENT ANALYSIS SUMMARY - REQUIREMENT TRACEABILITY MATRIX (RTM)

REQ ID	REQUIREMENT	TYPE	PRORITY	ACCEPTANCE CRITERIA
F1	User authentication and roles	Functional	High	Login works per role
F2	Incident and event logging	Functional	High	Create/update/close incidents; attach geo data
F3	Resource inventory and tracking	Functional	High	Add/edit resource; auto-update stock
F4	Evacuation center monitoring	Functional	High	Register centers; update occupancy data
F5	Situational dashboard and maps	Functional	High	Map pins, hazard layers, KPI dashboard
F6	Automated reporting	Functional	Medium	Generates reports such as PDFs
F7	Alert and notification system	Functional	Medium	Send SMS/push/email to selected subscriber groups; delivery status logged
N1	Availability	Non-Functional	High	Mobile app can collect incidents offline and sync when online; no data loss
N2	Security	Non-Functional	High	Data encryption and controlled access

N3	Usability	Non-Functional	Medium	UI text available in Tagalog & English

GATHERING REQUIREMENT CHALLENGES AND SUGGESTIONS

- **Stakeholder availability & competing priorities** — DRR staff are often busy with operations or drills.
  Suggestion: schedule short focused interviews, use async surveys, piggyback on existing meetings.
- **Uncommon knowledge and informal workarounds** — many important practices exist only in people's heads or WhatsApp groups.
  Suggestion: combine interviews with observation (ride-along, drill observation) and document analysis of forms/messages.
- **Conflicting requirements across stakeholders** — operations may want speed and simplicity; planners need detailed data for reporting [5](p. 16).
  Suggestion: use workshops to negotiate priorities and define Minimum Viable Product scope.
- **Limited IT capability and infrastructure** — LGUs may have limited network connectivity or legacy systems.
  Suggestion: design for offline mobile capture, lightweight web app, and clear integration plan.
- **Data quality and inconsistent formats** — hazard maps, inventories, and registries may be in different formats.
  Suggestion: define canonical data schemas early; include import scripts and data cleaning steps as part of the project.
- **Change resistance & training needs** — staff accustomed to paper/spreadsheets may resist a new system.
  Suggestion: involve end users early, run user acceptance testing, and plan hands-on training and quick reference guides.

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